Engine Model(s): PT6A

72-60-00

ACCESSORY GEARBOX ASSEMBLY - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-037	Enamel, Epoxy
PWC05-061	Cloth, Abrasive Coated Crocus
PWC05-101	Cloth, Abrasive
PWC11-014	Alcohol, Isopropyl
PWC13-001	Primer, Epoxy

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30046-52	Puller	
PWC30046-54	Puller	Post-SB1386
PWC30046-57	Puller	Alternative to PWC30052
PWC30051-01	Base	
PWC30051-02	Drift	
PWC30052	Puller	Alternative to PWC30046-57
PWC30054	Drift	Pre-SB1386

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Tool No.	Name	Application
PWC30075	Drift	
PWC30128-04	Puller	
PWC30274	Base	
PWC30373	Puller/Pusher	
PWC30415	Compressor	
PWC30499-50	Gage, spline wear	
PWC30675	Drift	Post-SB1386
PWC30854	Adapter	
PWC32275	Plate, Impeller	
PWC32396-400	Jackscrews	Replaces PWC42034 (No. 10-32)
PWC37088-001	Drift, Seal	
PWC37088-002	Drift, Seal	
PWC37088-003	Drift, Seal	
PWC42034	Jackscrews	Replaced by PWC32396-400
PWC50502	Compressor, AGB Centrifugal Impeller Ring	
PWC70838	Puller	
PWC70869	Drift	

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Removal/Installation

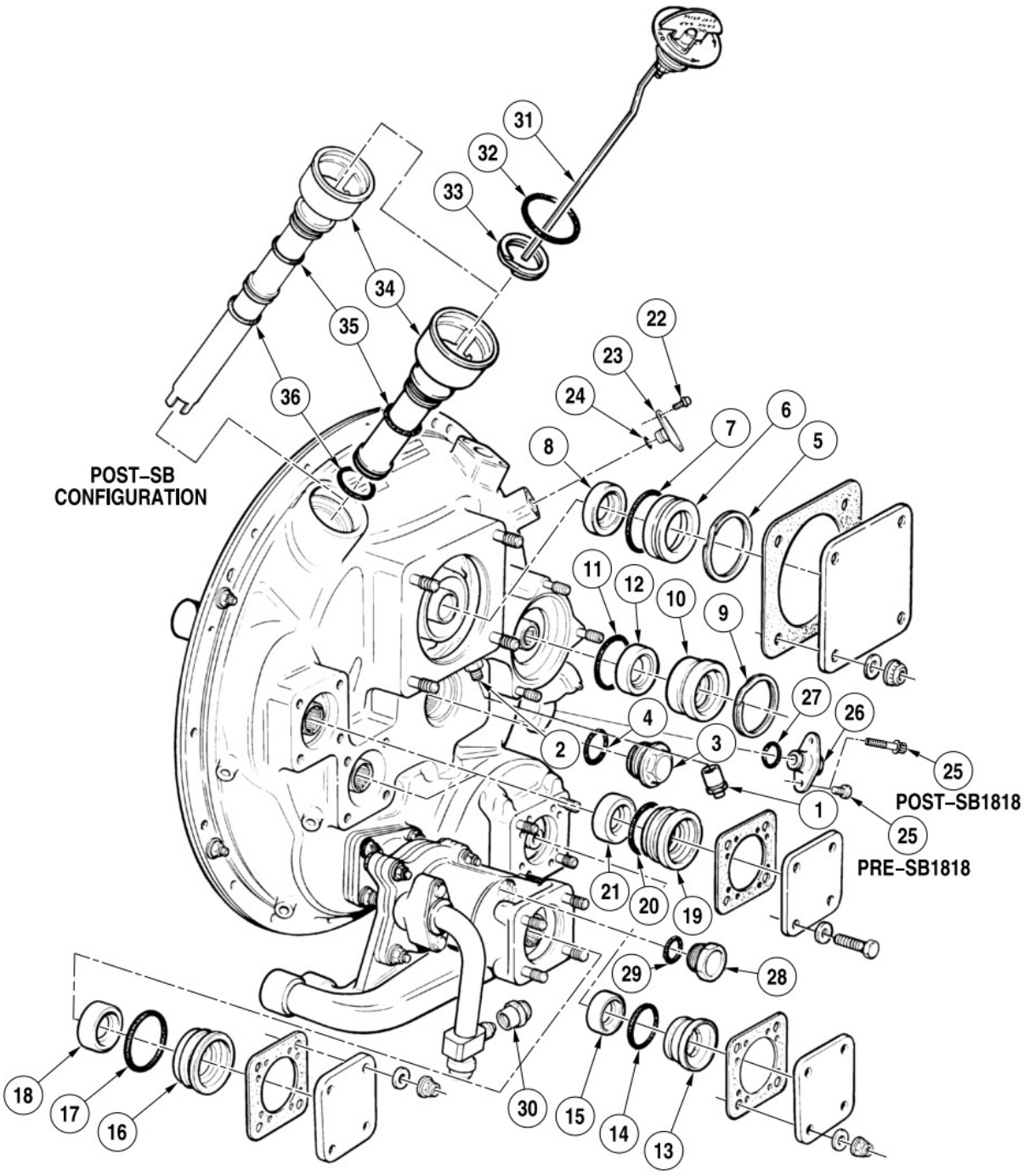
NOTE: Replacement of externally mounted driveshaft seals, oil pressure relief valve, oil filler cap/dipstick/gage assembly and oil filler tube is accomplished as a Line Maintenance Practice and detailed under Removal/Installation. Replacement of internally mounted driveshaft seals, centrifugal breather carbon seal and starter-generator gearshaft are done as a Heavy Maintenance Practice and detailed under Approved Repairs.

A. Removal of Accessory Drive Seals (Ref. Fig. 201)

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Figure 201 Removal of Oil Seal and Oil Seal Carrier (Typical)

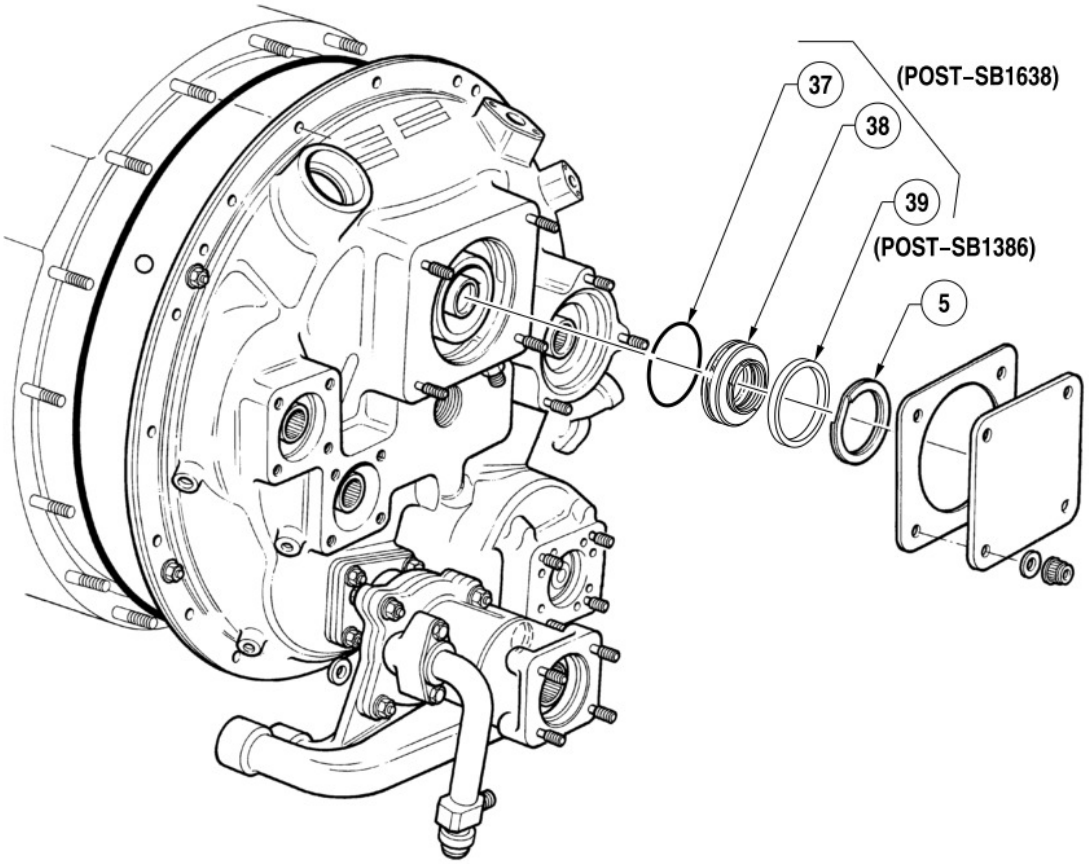
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1. Cap
2. Drain Adapter
3. Plug
4. Preformed Packing
5. Retaining Ring
6. Oil Seal Carrier (**Pre-SB1638**)
7. Preformed Packing (**Pre-SB1638**)
8. Seal (**Pre-SB1638**)
9. Retaining Ring
10. Oil Seal Carrier
11. Preformed Packing
12. Seal
13. Oil Seal Carrier
14. Preformed Packing
15. Seal
16. Oil Seal Retainer
17. Preformed Packing
18. Seal
19. Oil Seal Carrier
20. Preformed Packing
21. Seal
22. Bolt
23. Cover
24. Preformed Packing
25. Hex Bolt (**Pre-SB1818**) or Double Hex Bolt (**Post-SB1818** (**PT6A-21**))
26. Oil Temperature Bulb Adapter
27. Preformed Packing
28. Drain Plug
29. Preformed Packing
30. Cap
31. Filler Cap and Dipstick Assembly
32. Preformed Packing
33. Retaining Ring
34. Filler Tube (**Pre-SB1451** or **Post-SB1451**)
35. Preformed Packing
36. Preformed Packing
37. Packing (**Post-SB1638**)
38. Lip Seal (**Post-SB1638**)
39. Spacer (**Post-SB1386**/ **Post-SB1638** engines only)

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(1) Remove retaining ring (9, Fig. 201) from fuel pump mounting pads. Extract the oil seal carrier (10) using puller (PWC30046-54). Remove and discard the seal (12) and the preformed packing (11).

(2) Remove the tachometer-drive seal carrier (13) from the gearbox mount pad with the puller (PWC30052) or (PWC30046-57). Remove and discard the preformed packing (14) and the seal (15).

(3) Remove the oil seal retainer (16) from the scavenge pump housing with the puller (PWC30046-52). Remove and discard the preformed packing (17), seal (18) from the oil seal retainer.

(4) Remove starter generator oil seal carrier (Pre-SB1638) as follows:

(a) Remove retaining ring (5, Fig. 201) from the starter generator mounting pad. Extract seal carrier (6) from gearbox pad using puller (PWC30046-54).

(b) Remove and discard the preformed packing (7) and Seal (8).

(5) Remove starter generator lip-seal (Post-SB1638) as follows:

(a) Remove retaining ring (5, Fig. 201) from the starter generator mounting pad. Remove the spacer (39) from the lip seal (38) (For Post-SB1386 Engines only).

(b) Extract the lip seal (38) from the starter-generator mounting pad using puller (PWC70838).

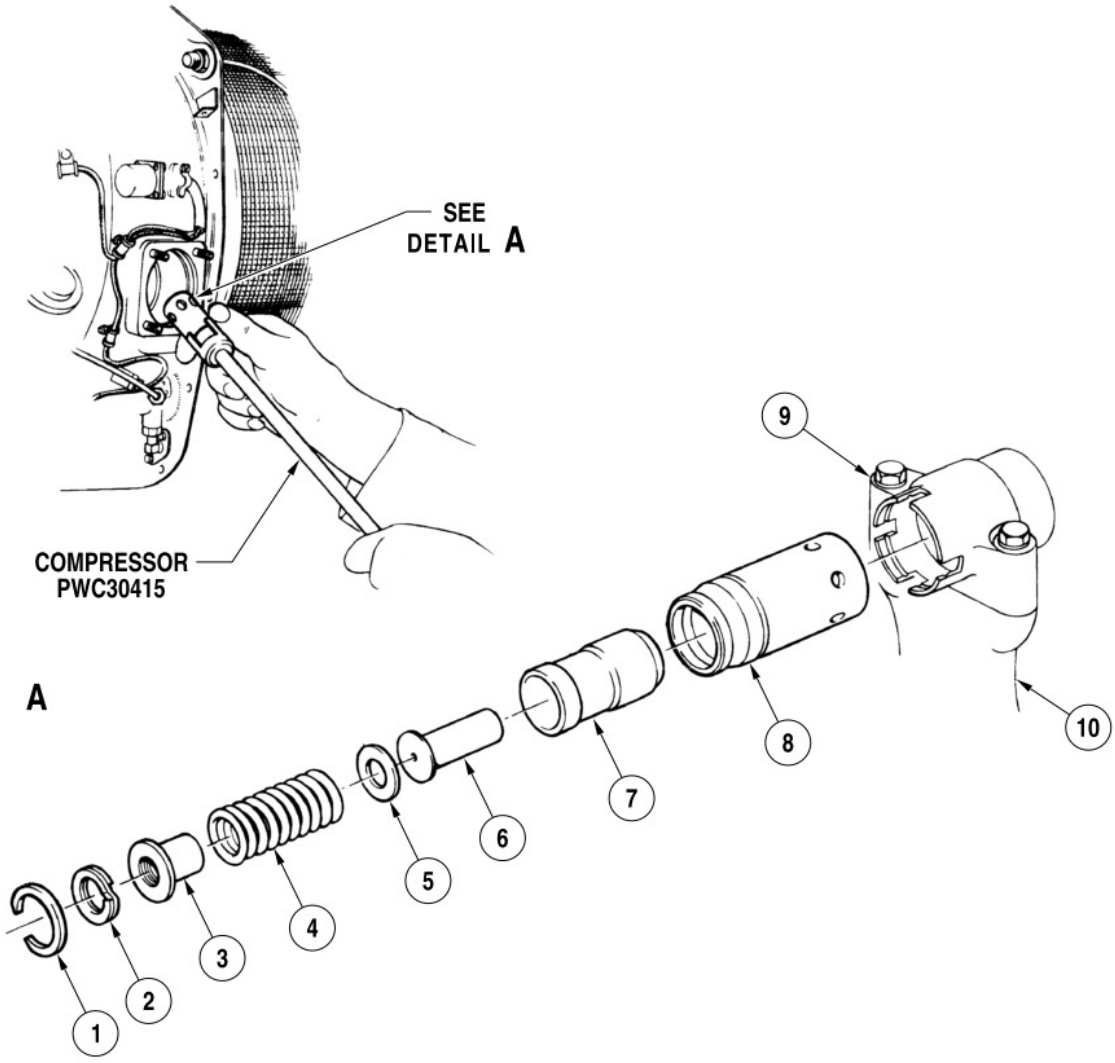
(c) Remove and discard the packing (37) and the lip seal (38).

(6) Remove two optional drive oil seal carriers (19) using puller (PWC30046-52). Remove and discard preformed packings (20) and seals (21).

B. Removal of Oil Pressure Relief Valve (Ref. Fig. 202)

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Figure 202 Removal/Installation of Oil Pressure Relief Valve



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1. Retaining Ring
2. Retaining Ring
3. Spring Seat
4. Spring
5. Spacer(s)(6 max.)
6. Guide
7. Relief Valve Piston
8. Sleeve
9. Relief Valve Housing
10. Oil Pump Body

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(1) Remove the oil filter element and housing (Ref. 79-20-02).

(2) Insert the oil relief valve compressor (PWC30415) through the oil filter housing port to collapse the retaining ring (1) securing the relief valve in the internal bore of the relief valve housing (9).

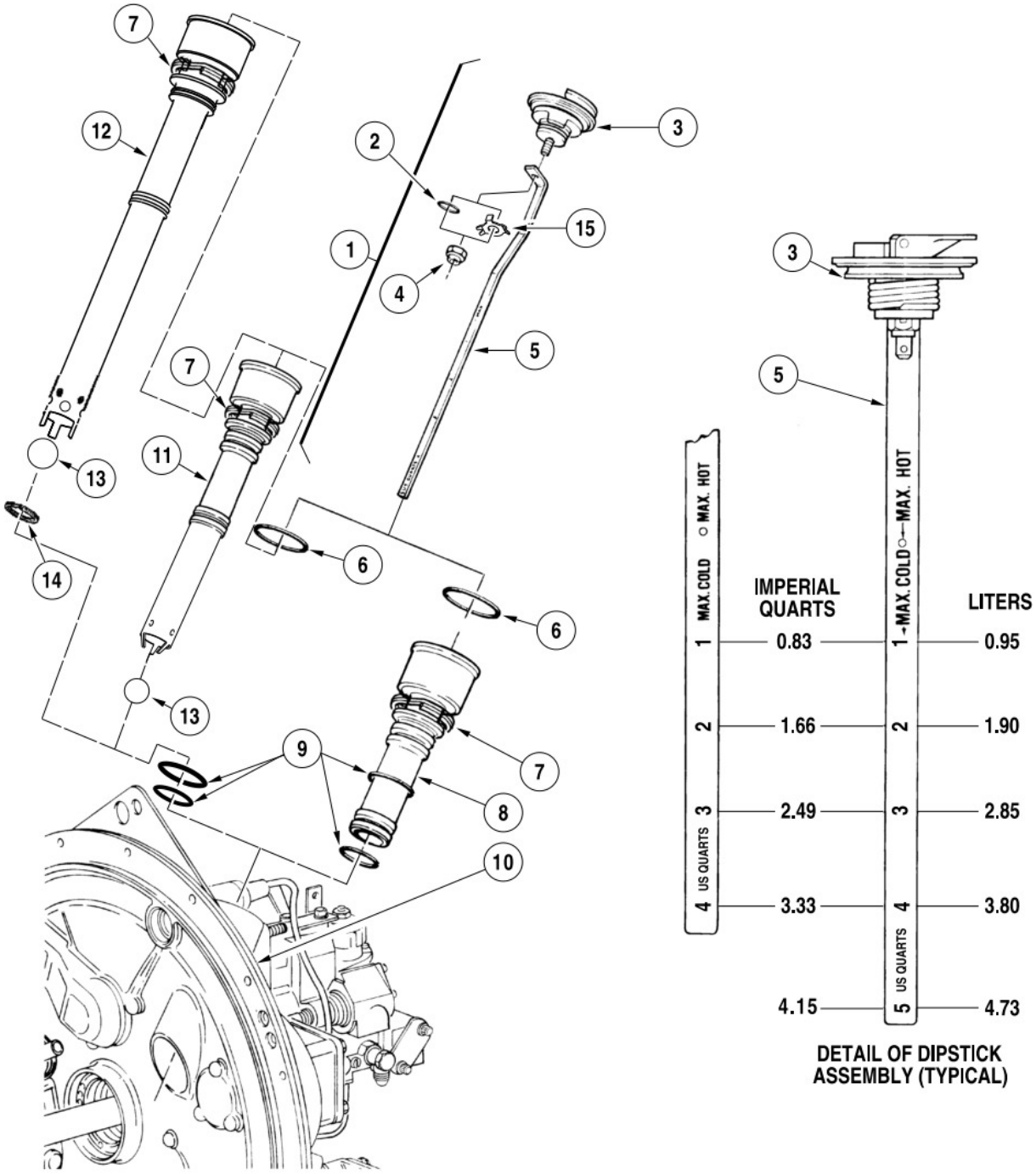
(3) Screw the shaft of the compressor into the spring seat (3) to compress the retaining ring (1). When the compressor shaft bottoms, pull on the shaft to extract the relief valve cartridge from the housing.

(4) Remove the retaining ring (2) from the sleeve (8). Extract the spring seat (3), spring (4), spacer(s) (5), guide (6) and relief valve piston (7) from the sleeve (8).

C. Removal of Oil Filler Cap Dipstick/Gage Assembly and Oil Filler Tube (Ref. Fig. 203)

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Figure 203 Removal/Installation of Oil Filler Cap, Dipstick/Gage Assembly and Oil Filler Tube



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1. Oil Filler Cap and Gage Assembly
2. Washer (Pre-SB1657)
3. Oil Filler Cap Spindle
4. Locknut
5. Dipstick
6. Preformed Packing
7. Retaining Ring
8. Filler Tube Assembly (Pre-SB1506/Pre-SB1744)
9. Preformed Packings
10. Accessory Gearbox Housing
11. Filler Tube Assembly (Post-SB1506/Pre-SB1659/Post-SB1744)
12. Filler Tube Assembly (Post-SB1659)
13. Plastic Ball (Post-SB1506/Post-SB1659/Post-SB1744)
14. Retaining Ring (Post-SB1659)
15. Keywasher (Post-SB1657)

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(1) Unlock and remove the oil filler cap and gage assembly (1) from the oil filler tube (8, 11 or 12). Remove the preformed packing (6).

(2) Disengage the retaining ring (7) attaching the oil filler tube (8, 11 or 12) to the accessory gearbox housing (10).

(3) Remove the oil filler tube (8, 11 or 12) from the accessory gearbox housing (10). Remove and discard the preformed packings (9). If the retaining ring (7) is unserviceable, remove it from the oil filler tube (8, 11 or 12).

(4) For Post-SB1659 Engines: Remove the retaining ring (14) and the plastic ball (13) from the oil filler tube (12).

(5) For **Post-SB1506/Post-SB1744** Engines: Remove the plastic ball (13) from the oil filler tube (11).

(6) When the dipstick requires replacement:

(a) Remove the locknut (4) and Pre-SB1657 washer (2) or Post-SB1657 keywasher that attaches the dipstick (5) to the underside of the oil filler cap body and remove the dipstick.

(b) Fit a new dipstick (5) on the oil filler cap spindle (3) and attach with Pre-SB1657 washer (2) or Post-SB1657 keywasher (15) and the locknut (4).

(7) Tighten the locknut 6 to 11 lb.in.

(8) For Post-SB1657 Engines: Lock the keywasher (15).

NOTE: The oil filler cap and gage assembly (1) are not repairable and should be returned to an approved overhaul facility.

D. Installation of Accessory Drive Seals (Ref. Fig. 201)

WARNING: PUT ON PROTECTIVE GLOVES WHEN YOU HANDLE HOT PARTS TO PREVENT PERSONAL INJURY.

(1) Put the oil seal carriers (10, 13, 16 and 19) in the oven at 141°C (285°F).

(2) Lubricate the oil seals (12, 15, 18 and 21) with engine oil (PWC03-001).

(3) Remove the oil seal carriers from the oven.

(4) Install each oil seal in their oil seal carrier using the drift (PWC37088-001), (PWC37088-002) or (PWC37088-003).

(5) Lubricate the preformed packing (11) with engine oil (PWC03-001), then install the preformed packing on the outer diameter of oil seal carrier (10).

(6) Lubricate the preformed packing (14) with engine oil (PWC03-001), then install the preformed packing on the outer diameter of oil seal carrier (13).

(7) Lubricate the preformed packing (17) with engine oil (PWC03-001), then install the preformed packing on the outer diameter of oil seal carrier (16).

(8) Lubricate the preformed packing (20) with engine oil (PWC03-001), then install the

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preformed packing on the outer diameter of oil seal carrier (19).

(9) Put the oil seal carriers in their boss. Carefully install each oil seal carrier into their boss with the drift (PWC30075).

(10) Secure the fuel control drive carrier assembly with retaining ring (9).

(11) Install gaskets and covers on respective bosses on gearbox housing with washers, self-locking nuts or bolts, as applicable. Torque nuts or bolts 65 to 85 lb. in.

(12) Install the starter generator oil seal carrier (Pre-SB1638) as follows:

(a) Heat the oil seal carrier (6) to 141°C (285°F) in an oven.

(b) Make the oil seal (8) moist with engine oil (PWC03-001).

(c) Remove the oil seal carrier from the oven.

(d) Install oil seal in its oil seal carrier using the drift (PWC37088-001), (PWC37088-002) or (PWC37088-003).

(e) Make the preformed packing (7) moist with engine oil (PWC03-001). Assemble preformed packing on the outside diameter of each oil seal carrier.

(f) Put the oil seal carrier in its boss. Carefully install each oil seal carrier into their boss with the drift (PWC30075).

(g) Secure starter-generator carrier assembly with retaining ring (5).

(h) Install gaskets and covers on respective bosses on gearbox housing with washers, self-locking nuts or bolts, as applicable. Torque nuts or bolts 65 to 85 lb.in.

(13) Install the starter generator lip seal (Post-SB1638) as follows:

NOTE: The lip seal (38) is supplied with a mandrel for installation. Do not remove the mandrel from the lip seal prior to installation.

(a) Apply engine oil (PWC03-001) to packing (37).

(b) Install the packing on the lip seal (38).

(c) Align the mandrel, of the lip seal, against the shaft in the boss of the gearbox housing.

(d) Push the lip seal into the shaft with drift (PWC70869).

(e) Discard the mandrel.

(f) For Post-SB1386 Engines, install the spacer (39) on the lip seal.

(g) Install the retaining ring (5) in the boss to secure the lip seal.

(14) Check for oil leakage (Ref. Chapter 71-00-00, ADJUSTMENT/TEST).

(15) Check for oil leakage after 5 to 10 hours of operation (Ref. Chapter 71-00-00, ADJUSTMENT/TEST, Post-Shutdown Check).

E. Installation of Oil Pressure Relief Valve (Ref. Fig. 202)

(1) Slide the relief valve piston (7) into the sleeve (8) followed by the guide (6), spacer(s) (5), spring (4) and spring seat (3). Secure the spring seat in the sleeve (8) with the retaining ring

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(2).

NOTE: A maximum of six spacers (5) may be used to establish the desired oil pressure. Each additional spacer will increase the oil pressure by three to five psi. Lapping of the valve is not recommended.

(2) Fit the retaining ring (1) into the exterior groove on the sleeve (8).

(3) Install the oil relief valve compressor into the threaded spring seat (3).

CAUTION: WHEN INSTALLING THE COMPRESSOR ON THE VALVE, MAKE SURE THAT THE GAP IN THE RETAINING RING IS SITUATED MIDWAY BETWEEN ANY TWO COMPRESSOR FINGERS. FAILURE TO DO SO WILL MAKE SUBSEQUENT VALVE REMOVAL DIFFICULT.

(4) Insert the relief valve, mounted on the valve compressor (Ref. Fig. 202, Detail A), through the filter housing port into the relief valve housing (9); be sure the valve is fully inserted.

(5) Slide the compressor body outward to release the retaining ring (1). Pull the shaft of the compressor outward until the retaining ring snaps into place. Unscrew the compressor shaft from the spring seat (3) and remove the compressor.

(6) Install the filter housing and filter element (Ref. 79-20-02).

F. Installation of Oil Filler Cap Dipstick/Gage Assembly, and Oil Filler Tube (Ref. Fig. 203)

(1) For Post-SB1659 Engines: Install the plastic ball (13) and the retaining ring (14) in the oil filler tube (12).

(2) For **Post-SB1506/Post-SB1744** Engines: Install the plastic ball (13) in the oil filler tube (11).

(3) Install the retaining ring (7) and the new lubricated preformed packings (9) on the oil filler tube (8, 11 or 12).

(4) Insert the oil filler tube (8, 11 or 12) into the accessory gearbox housing (10) aligning the notch in the flange of the tube with the dowel pin in the boss of the housing. Attach the oil tube (8, 11 or 12) with the retaining ring (7).

(5) Install the new preformed packing (6) on the oil filler cap and gage assembly (1) and insert into the oil filler tube (8, 11 or 12). With the oil filler cap correctly positioned and locked, no movement is allowed.

NOTE: For **Post-SB1506/Post-SB1744** Engines: When you install the oil filler tube assembly (11) have the ball type check valve, the applicable filler cap and gage assembly (1) must be installed.

(6) If the installed cap and gage assembly does show movement, remove and tighten the self-locking nut 6 to 11 lb.in. This should bottom out the nut. If the cap still shows movement when inserted and locked in the filler tube, replace with a new filler cap and gage assembly.

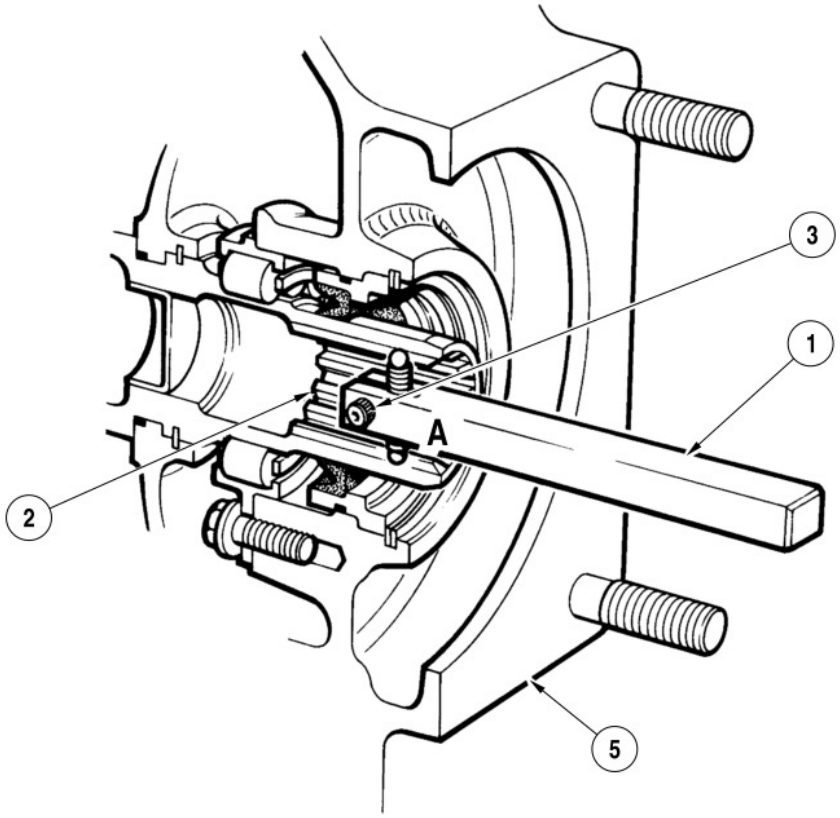
(7) If the required force to depress the lock tab by hand is excessive, remove the oil filler cap and back off the self-locking nut in increments of 1/4 turn at a time. Do not exceed 1/2 turn adjustment.

6. Inspection/Check

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A. Accessory Gearshaft Spline Wear Check (Ref. Fig. 204)

Figure 204 Accessory Gearshaft Spline Check



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1. Gearshaft Spline Wear Gage (PWC30499-50)
2. Gearshaft Splines
3. Gage Clamp Screw
4. Ball and Screw Assemblies
5. Starter-generator Mount Pad

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NOTE: The accessory gearshaft must be checked for spline wear at related component removal/installation (Ref. Aircraft Maintenance Manual).

(1) Release the clamp screw (3) of the gage (PWC30499-50) (1). Adjust the screw assemblies (4), then measure the Dim.A on the ball ends. The dimension must be 0.660 in. Tighten the clamp screws, then measure the Dim.A again.

(2) Tilt the gage (PWC30499-50) (1), then make sure that the outer end is up. Put the gage (PWC30499-50) into the shaft centerbore until the ball ends are approximately in middle from vertically opposite splines.

(3) Gradually release the gage, then make sure that the gage stays in position in the shaft. If the gage stays in the horizontal position, then the gearshaft is serviceable. If the gage extension falls below the horizontal position, then replace the gearshaft in the next 50 flight hours (Ref. Para. 8.D.).

B. Oil Filler Cap, Dipstick/Gage Assembly and Oil Filler Tube (Ref. Fig. 203)

(1) Examine the filler cap and level indicator assembly, and the oil filler tube for corrosion and signs of damage.

(2) For **Post-SB1506/Post-SB1744** Engines: Make sure that the ball moves freely in the check valve of the filler tube.

(3) Check the spring-loaded locking feature of the cap for serviceability. If the force required to depress/lift the handle is outside the limit (8 to 11 lbs.), remove the cap and tighten/loosen the self-locking nut to meet the required limit.

NOTE: Ensure the self-locking nut is securely fastened (refer to SB1637 - Inspection of Oil Tank Filler Cap).

(4) Check the gage blade for security and visible damage.

(5) When the cap is locked, there should be no movement. If movement is found, replace the cap.

C. Inspection of Scavenge Pump Inlet Screen

(1) Remove oil drain plug located on the AGB housing (Ref. Fig. 205).

NOTE: For engine equipped with an AGB chip detector, remove the chip detector.

(2) Collect drained oil in a clean container and check for any debris (Ref. Chap. 79-20-02, MAINTENANCE PRACTICES).

(3) Use a mirror and suitable light source (or borescope) and inspect the scavenge oil pump inlet screen for carbon or debris (Ref. Chap. 79-20-02, MAINTENANCE PRACTICES).

NOTE: Any foreign material found blocking the screen or contained in the oil should be identified. Look especially for paint flakes approximately 1/8 inch to 1/2 inch in diameter (Ref. Chap. 79-20-02, MAINTENANCE PRACTICES).

(4) If carbon or debris is present, clean the inlet screen (Ref. Cleaning/Painting, Fig. 205).

(5) Install the drain plug on the AGB housing (Ref. Chapter 72-00-00, SERVICING).

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NOTE: For engine equipped with an AGB chip detector, install the chip detector.

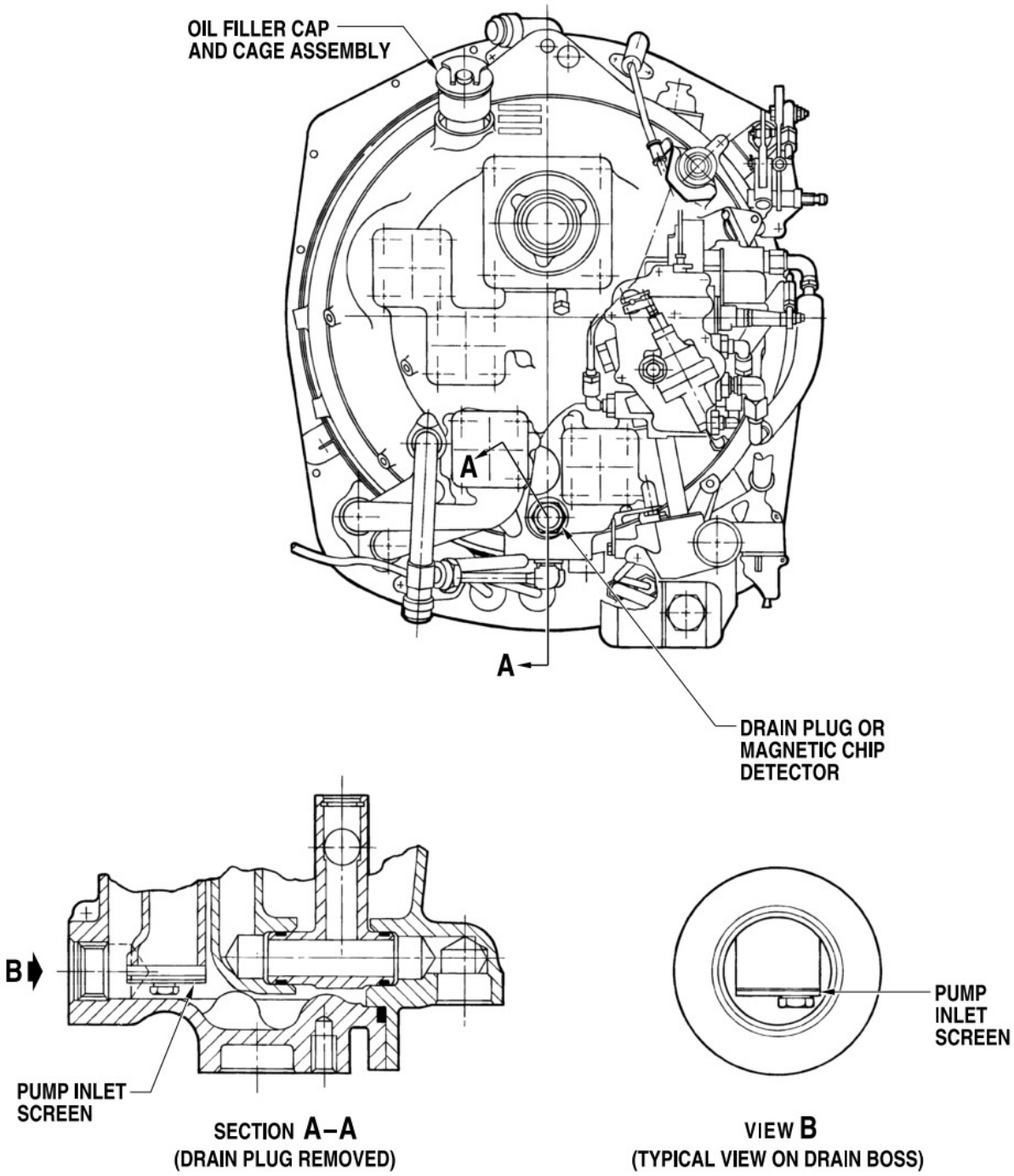
(6) Replenish the drained oil (if not contaminated by foreign material) in oil tank (Ref. Chapter 72-00-00, SERVICING).

7. Cleaning/Painting

A. Cleaning of Scavenge Oil Pump Inlet Screen (Ref. Fig. 205)

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Figure 205 Scavenge Pump Inlet Screen Inspection



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(1) If carbon is present on the screen mesh, loosen with a suitable small plastic scraper or similar nonmetallic scraper. Do not deform the screen mesh.

(2) Flush any carbon residue from the AGB as follows:

(a) Remove the oil filler cap (Ref. Removal/Installation).

(b) Remove the retaining ring and the oil filler tube (Ref. Removal/Installation)

(c) Place a suitable container under the AGB drain, and pour approximately 34 fluid ounces (one liter) of new engine oil (PWC03-001) heated to 122°F (50°C), into the AGB filler tube opening. Make sure oil does not enter oil tank.

(d) Let oil drain. Examine inlet screen and surrounding area to make sure carbon residue is removed. If not, repeat flushing procedure above.

(3) Install oil filler tube (Ref. Removal/Installation).

(4) Refill oil tank (Ref. Chapter 72-00-00, SERVICING)

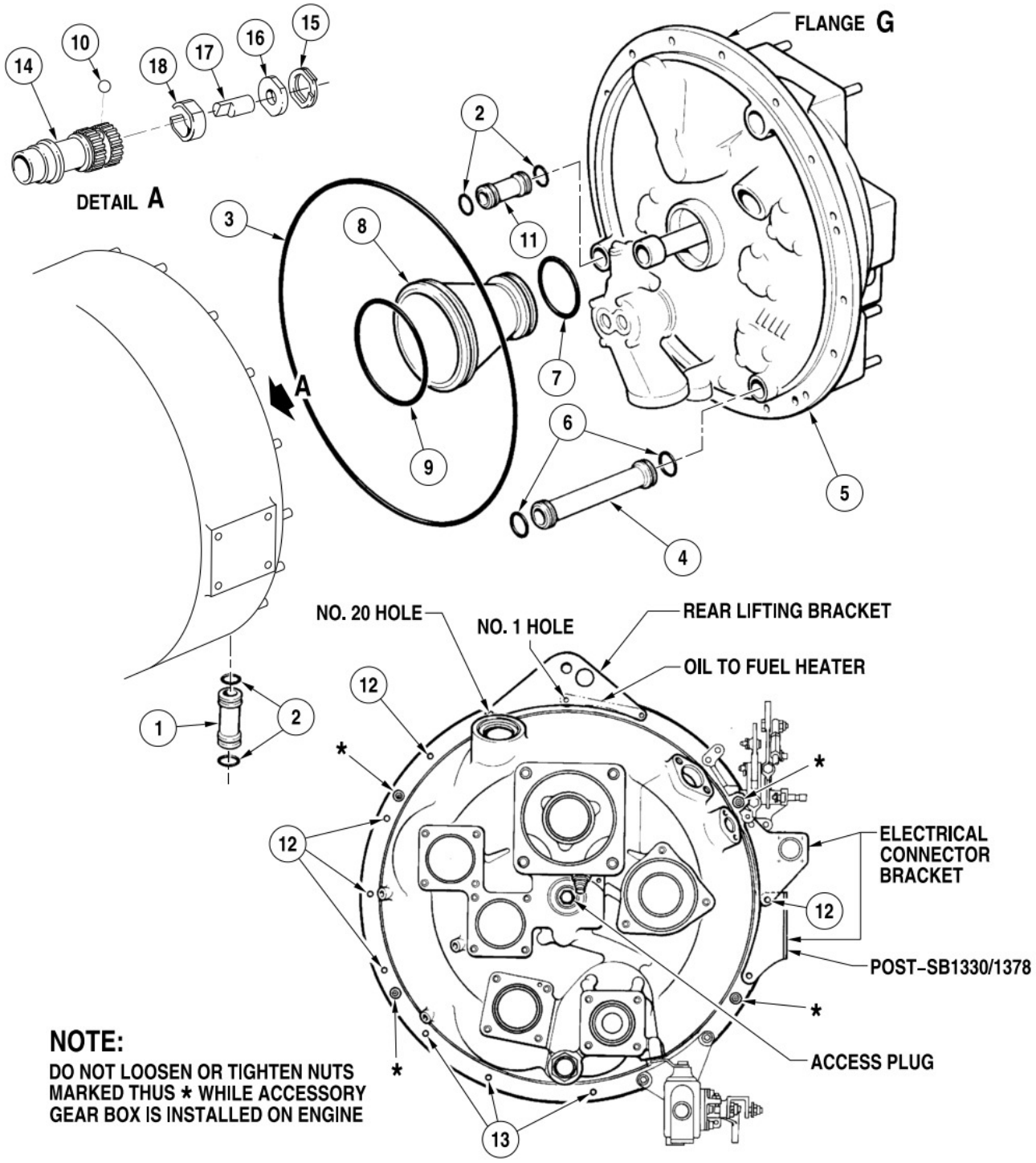
8. Approved Repairs

A. Removal of Accessory Gearbox from Compressor Inlet Case (Ref. Figs. 206 and 207)

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Figure 206 Removal/Installation of Accessory Gearbox

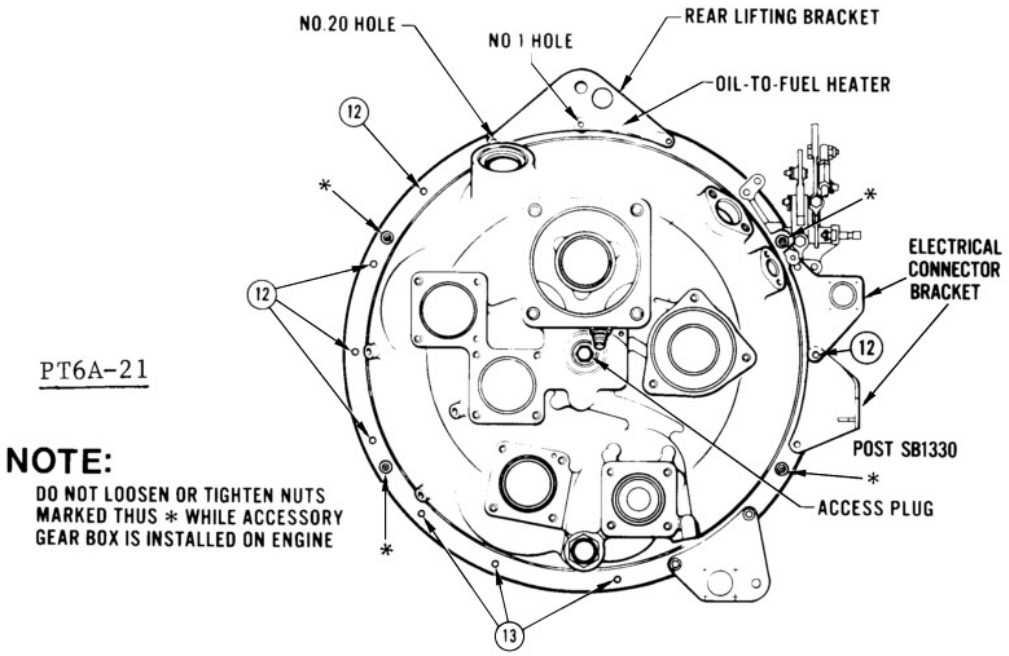
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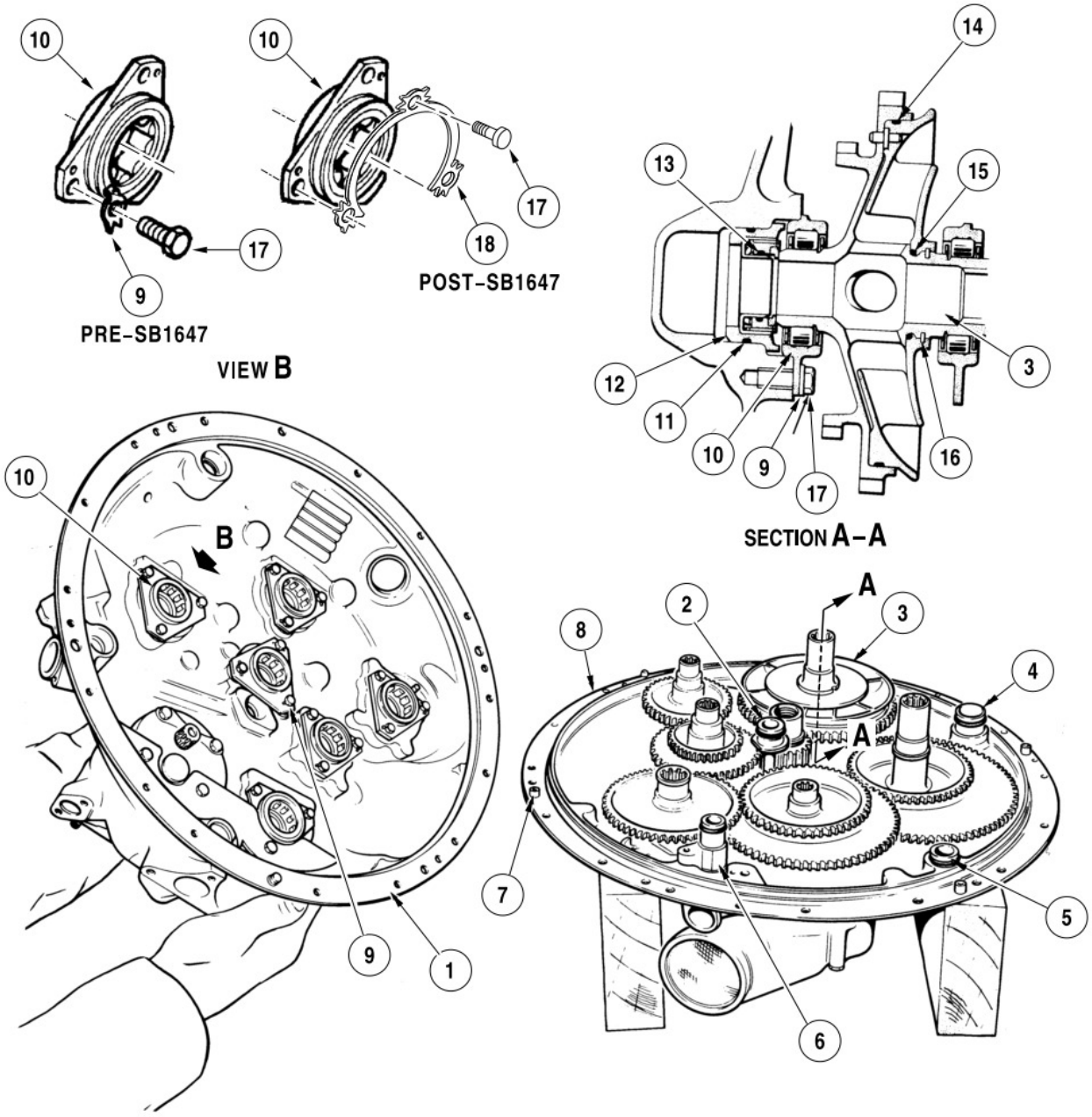
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1. No. 3 Bearing Internal Oil Transfer Tube
2. Preformed Packing
3. Preformed Packing
4. No. 1 Bearing Oil Transfer Tube
5. Accessory Gearbox Assembly
6. Preformed Packing
7. Center Tube Preformed Packing
8. Oil Tank Center Tube
9. Center Tube Preformed Packing
10. Lock Ball (Pre-SB1650)
11. No. 1 Bearing Oil Pressure Transfer Tube
12. Spacers Required
13. Washers Required
14. Compressor Rear Hub Coupling
15. Retaining Ring
16. Washer
17. Key
18. Expander Spring

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Figure 207 Removal/Installation of Accessory Gearbox Diaphragm



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1. Accessory Gearbox Housing
2. Center Flanged Transfer Tube and Oil Jet
3. Starter-generator Gearshaft and Centrifugal Impeller
4. Transfer Tube and Preformed Packing
5. Transfer Tube and Preformed Packing
6. No. 2 Scavenge Tube and Relief Valve Housing
7. Locating Dowels
8. Accessory Gearbox Diaphragm
9. Keywashers (Typical)(Pre-SB1647)
10. Flanged Roller Bearing (Typical)
11. Preformed Packing
12. Seal Carrier
13. Carbon Seal and Carrier
14. Preformed Packing
15. Preformed Packing
16. Retaining Ring
17. Bolt
18. Triple Keywashers (Typical)(Post-SB1647)

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(1) Drain the oil tank (Ref. 72-00-00, SERVICING).

(2) Remove the accessory gearbox drain plug and allow the residual oil to drain into a suitable container.

(3) If the engine is installed in the airframe, disconnect the following and remove as necessary:

(a) Connection(s) to the heated pneumatic tube electrical receptacle at Flange G.

(b) Connection to the overboard breather line.

(c) Connections to the oil pressure and temperature sensors and the Ng tachometer-generator.

(d) Connection to the starter-generator seal cavity drain.

(e) Connections to the fuel input at the oil-to-fuel heater, the FCU and the fuel pump.

(f) Connections and linkage to starting flow control (PT6A-27/-28 engines).

(g) Mechanical linkage connections to the FCU and reversing linkage.

(h) Starter-generator and such accessories as may be installed on the optional power takeoff mount pads.

(i) Connections to the airframe oil cooler.

(4) Remove the oil-to-fuel heater (Ref. 73-10-01).

(5) Remove the FCU and fuel pump as a complete assembly (Ref. 73-20-00 and 73-10-02).

NOTE: It is not necessary to separate the FCU from the fuel pump mating face.

(6) Remove starting flow control (PT6A-27/-28 engines) (Ref. 73-10-04).

(7) Disconnect the propeller reversing rear linkage mechanism at Flange G (Ref. 76-10-00), rear lifting bracket and electrical connector bracket.

(8) Disconnect and remove the propeller governor Py pneumatic line at the rear of the center fireseal bulkhead coupling (Ref. 73-10-08).

(9) Disconnect and remove the fuel pressure line(s) (Ref. 73-10-03).

(10) Remove the No. 2 bearing scavenge oil tube (Ref. 79-20-01) and remove the short oil transfer tube from the accessory gearbox diaphragm using the puller (PWC30128-04).

(11) Remove the oil dipstick and filler assembly and remove the oil filler tube from the accessory gearbox housing (Ref. Para. 5.C.).

(12) Remove the oil filter and filter housing from the compressor inlet case (Ref. 79-20-02).

(13) Remove the self-locking nuts securing the accessory gearbox housing and diaphragm assemblies to the compressor inlet case at Flange G.

NOTE: The accessory gearbox diaphragm is secured to the accessory gearbox housing by four countersunk screws and self-locking nuts. Assuming the nut at the 12 o'clock position to be No. 1, these nuts are 4th, 8th, 14th and 18th in a clockwise rotation and must not be removed at this stage.

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(14) Remove the starter-generator seal cavity drain plug from the accessory gearbox housing.

(15) Remove the access plug from the center boss adapter on the accessory gearbox housing, then remove the preformed packing from the plug.

(16) For Pre-SB1650 Engines: Find the location of the lock ball (10, Fig. 206) at the compressor rear hub coupling by viewing through the hole in the center boss. Turn the compressor until the notch on the washer at the rear hub is at the 3 o'clock position. This will make sure that the ball in the locking configuration is at the 12 o'clock position on the splined end of the rear hub.

(17) For Pre-SB1650 Engines: Put the accessory gearbox coupling shaft puller/pusher tool (PWC30373) through the center boss in the face of the accessory gearbox and screw into the center bore of the coupling shaft. Turn the knurled body into the center boss. Do not turn the puller/pusher T-handle when the tool is attached on the T-handle until the lock ball (10) is fully disengaged. Tighten the hexagon nut on the puller/pusher tool to remove any tension along the coupling shaft with hand.

(18) Install the accessory gearbox jackscrews (PWC42034) or (PWC32396-400) in the relevant locations on Flange G and carefully separate the accessory gearbox and diaphragm assembly from the studs on the compressor inlet case. Remove the preformed packing (3) from the diaphragm/inlet case flange. Remove the oil tank center tube (8) and preformed packings (7 and 9).

NOTE: For Pre-SB1650 Engines: If the lock ball (10) has dropped out of the locking arrangement during the removal of the accessory gearbox, retrieve the ball from the No. 1 bearing housing area or the oil scavenge passage in the compressor inlet case.

(19) For Pre-SB1650 Engines: Remove the lock ball (10) from the locking configuration on the top of the compressor rotor hub coupling (14), and the retaining ring (15), washer (16), key (17) and expander spring (18) from the inside of the hub. Discard the lock ball and expander spring.

NOTE: If the lock ball has dropped out of the locking configuration during the removal of the accessory gearbox, retrieve the ball from the No. 1 bearing housing area or the oil scavenge passage in the compressor inlet case.

(20) Support the accessory gearbox housing (1, Fig. 207) and diaphragm assembly (8) on suitable wooden blocks with diaphragm face down on the work bench.

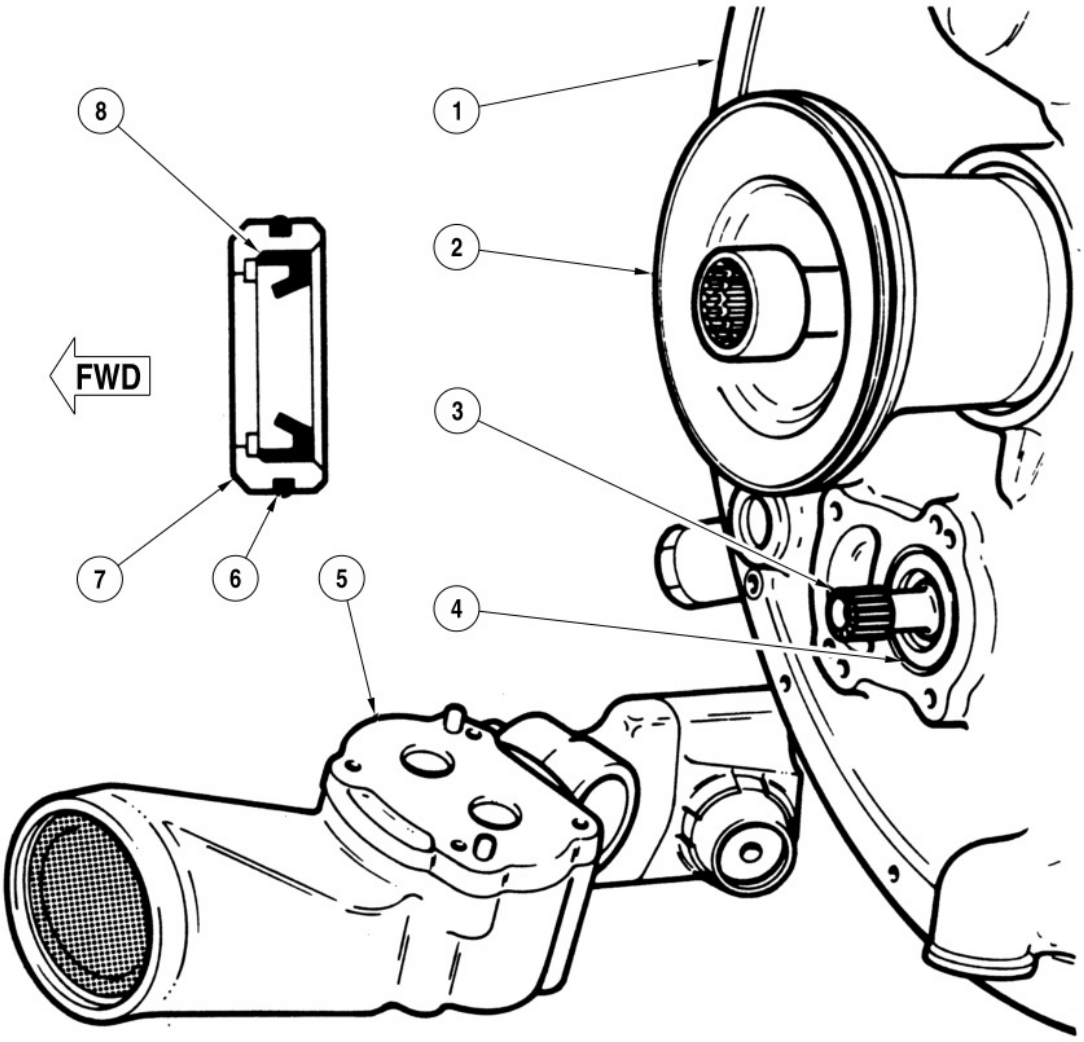
(21) Remove the puller/pusher tool. Remove the self-locking nuts, washers and screws fastening the gearbox housing to the diaphragm. Lift the housing from the diaphragm; use a soft-faced hammer to separate, if necessary. Remove the preformed packing.

(22) Remove the transfer tubes (2, 4 and 5) and remove the preformed packings from the tubes. Remove the preformed packing(s), as applicable, from the No. 2 bearing scavenge tube and relief valve housing assembly (6).

B. Replacement of Oil Pump Seal (Ref. Fig. 208)

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Figure 208 Removal/Installation of Oil Pump Drive Seal



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1. Accessory Gearbox Diaphragm
2. Oil Tank Center Tube
3. Oil Pump Driveshaft
4. Oil Seal and Carrier
5. Oil Pump
6. Preformed Packing
7. Oil Seal Carrier
8. Oil Seal

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(1) Remove the accessory gearbox housing and diaphragm assembly from the engine (Ref. Removal/Installation).

(2) Remove oil pump (5).

(3) Remove and install the replacement oil pump oil seal (Ref. Paras. 5.A. and 5.D.).

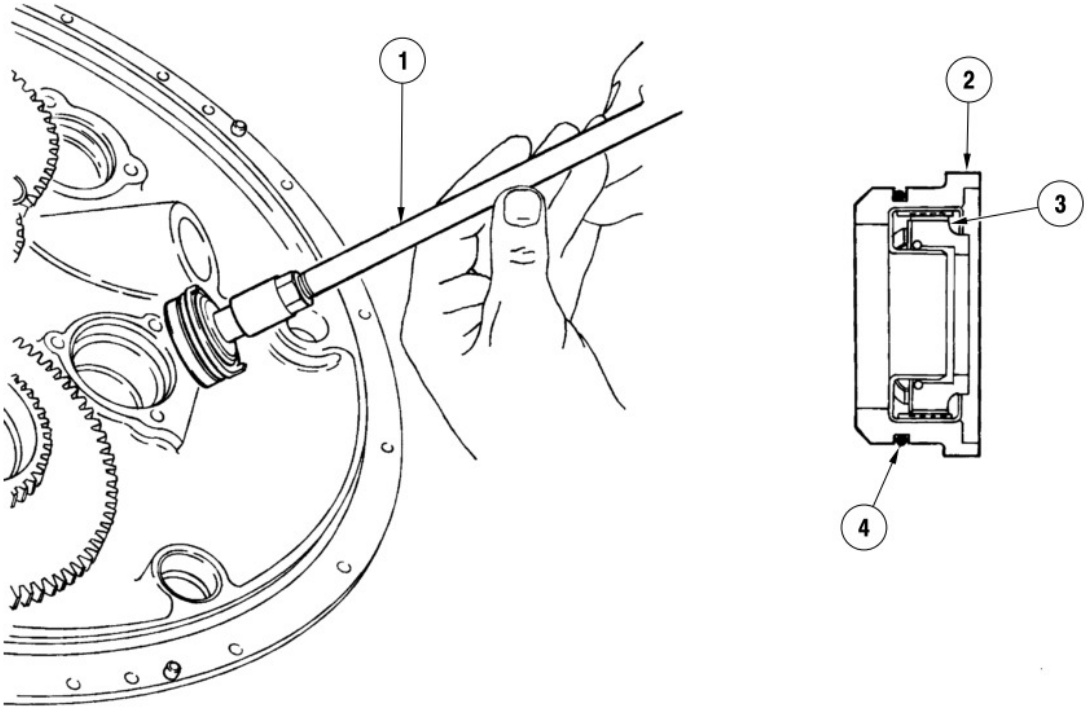
NOTE: The oil seal is fitted from the non-pressure side.

(4) After the installation of the oil pump (5) on the gearbox diaphragm, tighten the four attachment bolts 32 to 36 lb.in. and fasten with lockwire.

C. Replacement of Starter-generator Centrifugal Breather Gearshaft Carbon Seal (Ref. Figs. 207 and 209)

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Figure 209 Replacement of Starter-generator Centrifugal Breather Gearshaft Carbon Seal



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1. Carbon Seal Carrier Puller (PWC30052 or PWC30046-57)
2. Carbon Seal Carrier
3. Carbon Seal and Casing
4. Preformed Packing

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(1) Remove the accessory gearbox housing and diaphragm assembly from the engine and remove the housing from the diaphragm (Ref. Para. 8.A.).

(2) Remove the starter-generator/centrifugal breather gearshaft (3, Fig. 207) from the gearbox housing (1).

(3) For Pre-SB1647 Engines: Remove the bolts (17) and the key washers (9) that attach the flanged bearing (10) to the diaphragm. Remove the bearing.

(4) For Post-SB1647 Engines: Remove the bolts (17) and the triple key washers (18) that attach the flanged bearing (10) to the diaphragm. Remove the bearing.

(5) Remove the seal and the seal carrier (2, Fig. 209) with the puller (1, (PWC30052) or PWC30046-57). Remove and discard the preformed packing (4).

(6) Remove the carbon seal and casing (3) from the carrier (2) using the carbon seal base (PWC30051-01) and drift (PWC30051-02).

(7) Immerse the new carbon seal in engine oil (PWC03-001) and move the seal in and out within the seal casing to make sure there is freedom of movement.

WARNING: PUT ON PROTECTIVE GLOVES WHEN HANDLING THE HOT PARTS TO PREVENT PERSONAL INJURY.

(8) Put the seal carrier in an oven at 141 °C (285 °F) for 15 minutes minimum.

WARNING: PUT ON PROTECTIVE GLOVES WHEN HANDLING THE HOT PARTS TO PREVENT PERSONAL INJURY.

(9) For **Pre-SB1386** Engines: Remove the carrier from the oven. Position the replacement carbon seal in the carrier (Ref. Fig. 209), then use drift (PWC30054) to fully install the seal in the carrier.

WARNING: PUT ON PROTECTIVE GLOVES WHEN HANDLING THE HOT PARTS TO PREVENT PERSONAL INJURY.

(10) For **Post-SB1386** Engines: Remove the carrier from the oven. Position the replacement carbon seal in the carrier (Ref. Fig. 209), then use drift (PWC30675) to fully install the seal in the carrier.

(11) Install a new lubricated preformed packing (4) on the seal carrier and press the carrier into the boss on the diaphragm.

(12) For Pre-SB1647 Engines: Install the flanged bearing (10, Fig. 207) on the boss and attach with three bolts (17) and keywashers (9). Tighten the bolts 36 to 40 lb.in., and lock the keywashers.

(13) For Post-SB1647 Engines: Install the flanged bearing (10, Fig. 207) on the boss and attach with three bolts (17) and triple keywashers (18). Tighten the bolts 36 to 40 lb.in., and lock the keywashers.

(14) Install the starter-generator/centrifugal breather gearshaft (3) in the gearbox housing.

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CAUTION: IF, FOR ANY REASON, OTHER ACCESSORY DRIVES WERE REMOVED DURING THE PROCEDURE, EXERCISE CARE NOT TO INTERCHANGE THE EXTERNAL SCAVENGE PUMP GEARSHAFT (NO. 4) WITH THE HYDRAULIC PUMP GEARSHAFT (NO. 6). THESE GEARSHAFTS ARE SIMILAR IN APPEARANCE AND MUST BE IDENTIFIED BY PART NUMBER.

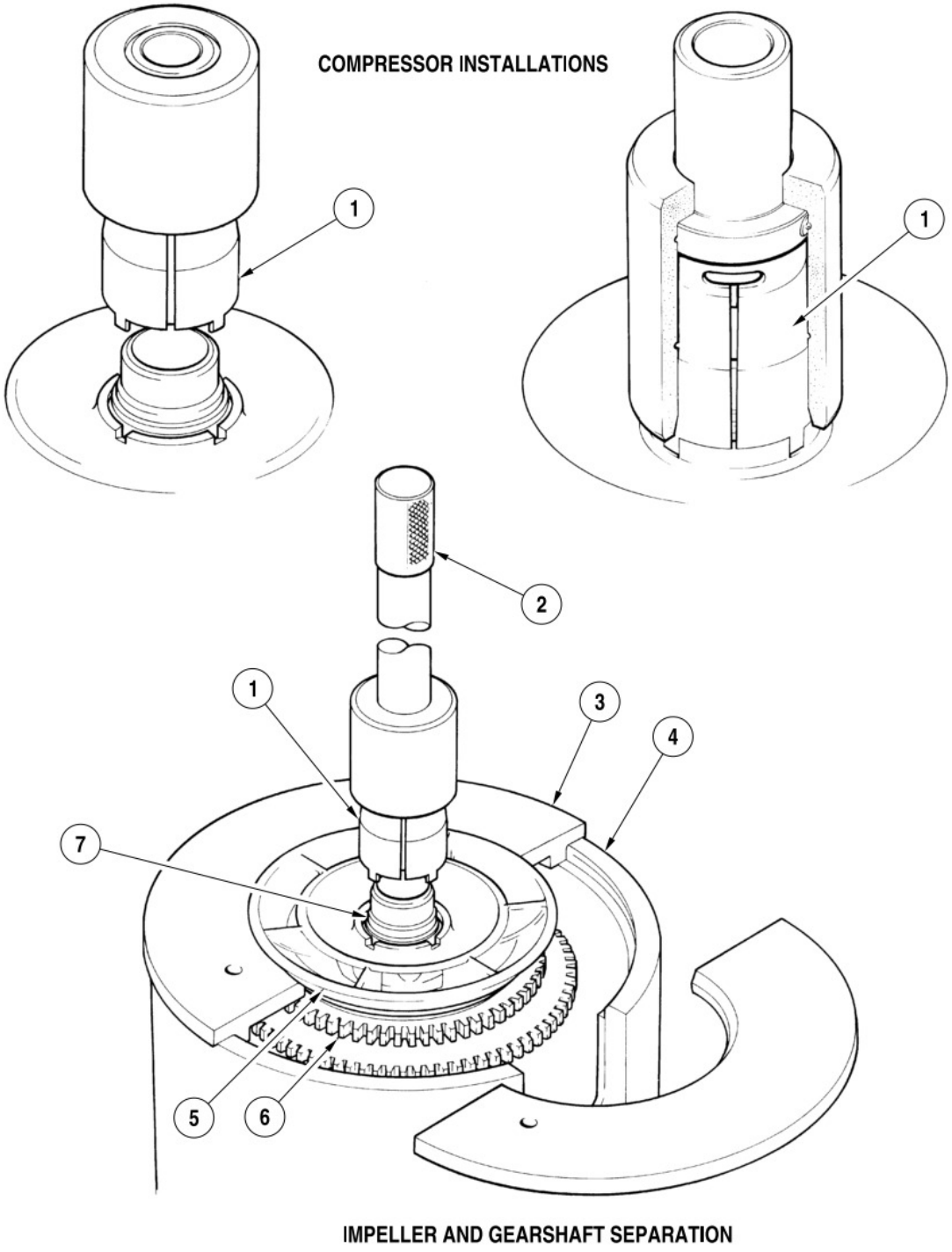
(15) Install the accessory gearbox housing on the diaphragm.

(16) Install the accessory gearbox assembly on the engine (Ref. Para. 5.E.).

D. Replacement of Starter-generator Gearshaft

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Figure 210 Replacement of Starter-generator Gearshaft



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1. Compressor (PWC50502)
2. Drift (PWC30854)
3. Split Plate (PWC32275)
4. Base (PWC30274)
5. Centrifugal Impeller
6. Starter-generator Gearshaft
7. Retaining Ring

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(1) Disassembly (Ref. Figs. 207 and 210):

(a) Remove the accessory gearbox housing and diaphragm assembly from the engine, and remove the housing from the diaphragm (Ref. Para. 5.A.).

(b) Remove the starter-generator/centrifugal breather gearshaft (3, Fig. 207) from the gearbox diaphragm (8).

(c) Locate the split plate (3, Fig. 210) as shown, between the centrifugal impeller (5) and the starter-generator gearshaft (6).

(d) Position the assembly on the base (4).

(e) Install the compressor (1) on the impeller (5) and mate the lugs of the compressor with the slots in the impeller.

CAUTION: SEPARATION OF THE GEARSHAFT AND IMPELLER SHOULD NOT REQUIRE EXCESSIVE FORCE. IF DIFFICULTY IS EXPERIENCED, CHECK THAT THE RETAINING RING IS COMPLETELY CLEAR OF THE GROOVE IN THE IMPELLER.

(f) Using the drift (2) inserted in the compressor, separate the assembly.

(g) Remove the split plate from the impeller and the gearshaft from the bottom of the base.

(h) Remove the retaining ring (16, Fig. 207) and preformed packing (15) from the gearshaft.

(i) Remove the preformed packing (14) from the impeller.

(2) Assembly (Ref. Fig. 207):

(a) Install the preformed packing (14) on the impeller.

(b) Install the preformed packing (15) on the gearshaft.

(c) Install the retaining ring (16) in the groove in the gearshaft.

NOTE: Check for the security of the shoulder pins before mating the parts.

(d) Position the impeller on the gearshaft; align the shoulder pins in the impeller with the holes in the gearshaft and press the parts firmly together by hand to engage the retaining ring. Check that the parts are locked together.

(e) Install the starter-generator/centrifugal breather gearshaft on the diaphragm (8) by mating with the related gearshafts.

(f) Assemble the accessory gearbox housing on the diaphragm and install the assembly on the engine (Ref. Para. 8.E.).

E. Installation of Accessory Gearbox on Compressor Inlet Case (Ref. Figs. 206 and 207)

(1) Install new preformed packings on the three transfer tubes (2, 4 and 5, Fig. 207) and insert the tubes into the respective bosses on the accessory gearbox diaphragm (8).

(2) Install new preformed packing(s), as applicable, on the No. 2 bearing scavenge tube and relief valve housing assembly (6).

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CAUTION: FOR Post-SB1638 ENGINES, IF THE AGB HOUSING IS SEPARATED FROM THE DIAPHRAGM WITHOUT REMOVING THE LIP SEAL ON THE STARTER-GENERATOR MOUNTING PAD, THE LIP SEAL MUST BE REMOVED AND DISCARDED BEFORE INSTALLING THE HOUSING ON THE DIAPHRAGM. IT IS NOT POSSIBLE TO RE-MATE THE AGB WITH THE LIP SEAL INSTALLED. FOR INSTALLATION OF NEW LIP SEAL, REFER PARA. 5.D..

(3) Install a new preformed packing on the rear flange of the accessory gearbox diaphragm (8) and position the housing (1) over the diaphragm. The lead-in chamfer on the gearshafts permits easy entry of the shafts into the respective bearings and oil seals.

NOTE: Make sure that the flanged transfer tube (2) is fully installed in the center boss of the diaphragm (8), with the flat aligned with the center bearing flange before positioning the housing (1).

(4) Seat the housing (1) on the diaphragm (8), making sure of the proper engagement of the locating dowels (7), without using force.

(5) Locate the four countersunk screws, washers and self-locking nuts in the relevant locations on Flange G. Screws are located at the 4th, 8th, 14th and 18th positions from the top center in a clockwise direction. Tighten the nuts 32 to 36 lb.in.

(6) Install new expander spring (18), key (17), and washer (16) into compressor rear hub coupling (14), and secure with retaining ring (15).

(7) Before raising the gearbox to the normal position, install the accessory gearbox driveshaft puller/pusher (PWC30373) through the center boss of the housing to engage the thread of the driveshaft. Screw the body of the pusher into the housing.

(8) Install a new preformed packing (3) on the front flange of the diaphragm and install the oil tank center tube (8) with new preformed packings (7 and 9) into the boss at the center bore of the diaphragm.

(9) For Pre-SB1650 Engines: Slowly turn the compressor until the hole in the splined end of the compressor rear hub coupling is at the 12 o'clock position. For Pre-SB1104/Post-SB1134 PT6A-27/-28 engines and all PT6A-21 Engines: Layer the lock ball (10) with engine oil (PWC03-001). Install the lock ball in the hole.

NOTE: For Post-SB1650 Engines: The AGB drive gearshaft is replaced with a new one that has a double shouldered feature which eliminates the necessity for a bearing-ball locking system.

(10) Position the accessory gearbox housing and diaphragm assembly on the studs of the compressor inlet case aligning the driveshaft with the compressor rear hub coupling.

NOTE: Make sure all tubes mate with the corresponding bosses in the inlet case, and that the external oil tubes mate with the scavenge pump elbows.

(11) For Pre-SB1650 Engines: Make sure that the lock ball (10) is seated in the hole in the compressor rear hub and then apply hand pressure on the puller/pusher to engage the ball.

NOTE: The correct engagement of the lock ball will be indicated by the end-float along the driveshaft being reduced to a small fore and aft movement.

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(12) Remove the driveshaft puller/pusher from the engine. Install a new preformed packing on the access plug and screw the plug into the adapter on the gearbox housing. Tighten the plug 150 to 160 lb.in. and fasten with lockwire.

CAUTION: DO NOT OVER-TIGHTEN THE ADAPTER. THE TAPERED PIPE THREADED END ENTERS THE HOUSING CASTING AND OVERTIGHTENING MAY STRIP THE THREADS.

(13) Install the starter-generator seal cavity drain adapter and plug on the drain adapter boss at the bottom of the starter-generator mounting pad. Tighten the adapter and plug 30 to 40 lb.in. Fasten the adapter, plug and center plug to the locking eye on the gearbox housing with lockwire.

(14) Install the washers (13) over the short studs, and spacers (12) over the long studs, at Flange G and install self-locking nuts. Tighten the nuts 32 to 36 lb.in.

NOTE: Washers are not required at the rear lifting bracket.

(15) Install the oil filter housing and filter (Ref. 79-20-02, OIL FILTER, HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES).

(16) Install the oil dipstick/gage assembly and oil filler tube in the accessory gearbox housing (Ref. Para. 5.F.).

(17) Fit new preformed packings (2) on the short oil transfer tube (1). Install the tube in the compressor inlet case and connect the No. 2 bearing scavenge oil tube to the boss on the case (Ref. 79-20-02).

(18) Secure loop clamps, installed on ignition leads to similar loop clamp on No. 2 bearing scavenge oil tube with bolt and self-locking nut. Tighten nuts and torque 32 to 36 lb.in.

(19) Connect the power turbine governor Py pneumatic line to the rear of the bulkhead coupling at the center fireseal (Ref. 73-10-08).

(20) Install the rear section(s) of the compressor discharge pressure pneumatic line (Ref. 73-10-07), and connect the electrical leads as necessary to the electrical receptacle on the bracket on Flange G.

(21) Reconnect the rear section of the propeller reversing linkage mechanism at Flange G (Ref. 76-10-00).

(22) Install the FCU and fuel pump as a complete assembly on the FCU mount pad (Ref. 73-20-00 and 73-10-02).

(23) Install starting flow control unit (PT6A-27/-28) (Ref. 73-10-04).

(24) Install the oil-to-fuel heater (Ref. 73-10-01).

(25) Connect the fuel pressure line(s) (Ref. 73-10-03).

(26) Install the accessory gearbox drain plug at the 6 o'clock position on the gearbox housing. Tighten the plug 215 to 240 lb.in. and fasten with lockwire.

(27) Fill the oil tank (Ref. 72-00-00, SERVICING).

(28) Pressure test the pneumatic system (Ref. 73-10-07).

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(29) Install and connect the following, as necessary (Ref. Aircraft Maintenance Manual):

(a) Connections to the airframe oil cooler.

(b) Starter-generator and accessories on the optional power takeoff mount pads.

(c) Mechanical linkage connections to the FCU and reversing linkage.

(d) Connections to the starting control, if applicable.

(e) Connections to the fuel input at the oil-to-fuel heater, the FCU and the fuel pump.

(f) Connections to the starter-generator seal cavity drain.

(g) Connections to the oil pressure and temperature sensors and the Ng tachometer.

(h) Connection to the overboard breather line.

(i) Connection to the heated pneumatic tube electrical receptacle at Flange G.

(30) Test the engine (Ref. 71-00-00).

F. Repair of Corrosion - Accessory Gearbox Housing

NOTE: Corrosion must not be deeper than 0.010 in. and must not cover an area greater than 10 percent of the total surface of the AGB housing.

(1) Apply a suitable covering material over the accessories, linkages, compressor section and around the area to be repaired.

(2) Clean the surface of the area to be repaired with a swab soaked in isopropyl alcohol (PWC11-014) and crocus cloth (PWC05-061).

(3) Remove all traces of corrosion (magnesium oxide) using a suitable steel brush, crocus cloth (PWC05-061), abrasive cloth (PWC05-101), file or grit paper. Remove all traces of debris using a vacuum cleaner.

(4) Flush the area with clean water at room temperature.

(5) Dry the area with clean compressed air at 29 psig.

WARNING: REFER TO THE MATERIAL SAFETY DATA SHEETS BEFORE YOU USE THESE MATERIALS FOR INFORMATION SUCH AS HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL CHARACTERISTICS, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES. SOME OF THESE MATERIALS CAN BE DANGEROUS. YOU CAN GET THE DATA SHEETS FROM THE MANUFACTURERS OR THE SUPPLIERS OF THESE MATERIALS.

(6) Prepare a chrome pickle solution (Ref. Chap. 70-00-00, MAINTENANCE PRACTICES, Touch-Up Solution).

(7) Using a swab or brush, apply the chrome pickle solution, at a temperature of 17° to 29°C (55° to 85°F), to the prepared surface for 30 to 45 seconds.

NOTE: Repeat the application frequently to make sure that the affected surface is continually wet with the solution.

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(8) Swab the area with clean water until successive swabs are no longer stained yellow.

(9) Dry the area with local heating (air dry or a heat gun at a low setting).

(10) Clean the affected area with a rag soaked in clean water.

(11) Dry the area with clean, compressed air at 29 psig.

(12) Apply two coats of primer (PWC13-001).

NOTE: The primer can be diluted with 10% solvent.

(13) Allow the primer to air dry for eight hours before applying enamel paint. Use compressed air at 29 psig to accelerate the drying time.

(14) Apply three to four coats of enamel (PWC05-037) to the primed surface. Allow the surface of the enamel to become tacky (approximately 15 minutes) between each coat. The final coat of enamel must dry for 24 hours before returning the engine to service.

NOTE: Drying time for the primer and paint can be reduced with the use of a heat gun at a low setting. Refer to the manufacturers instructions.

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